

Clause 172 was added to IEEE Std 802.3-2022 by IEEE Std 802.3df-2024:

172. Physical Coding Sublayer (PCS), type 800GBASE-R

172.1 Overview

172.1.3 Summary of functions

Change the text in 172.1.3 as follows:

The PCS service interface is the 800 Gb/s Media Independent Interface (800GMII), which is defined in Clause 170. The 800GMII provides a uniform interface to the Reconciliation Sublayer for all 800 Gb/s PHY implementations.

The 800GBASE-R PCS is based on the 400GBASE-R PCS defined in Clause 119.

The 800GBASE-R PCS provides all of the services required by the 800GMII, including the following:

- a) Encoding (decoding) of eight 800GMII data octets to (from) 66-bit blocks (64B/66B)
- b) Transcoding from (to) 66-bit blocks to (from) 257-bit blocks (256B/257B)
- c) Scrambling (descrambling) the 257-bit transcoded blocks
- d) Alignment marker insertion and removal
- e) Reed-Solomon encoding (decoding) the scrambled 257-bit blocks
- f) Distribution of Reed-Solomon encoded data to (from) PCS lanes
- g) Alignment marker lock, lane deskew and reordering of received PCS lanes
- h) Transferring encoded data to (from) the PMA sublayer below the PCS
- i) Compensation for any rate differences caused by the insertion or deletion of alignment markers or due to any rate difference between the 800GMII and PMA the sublayer below the PCS through the insertion or deletion of idle control characters

172.1.5 Inter-sublayer interfaces

Change the text in 172.1.5 as follows:

The upper interface of the PCS connects to the Reconciliation Sublayer or a PHY 800GXS through the 800GMII PCS service interface. The lower interface of the PCS connects to the a PMA or a FEC sublayer, to support a PMD. The lower interface of the 800GBASE-R PCS has a nominal rate at the PMA service interface of 26.5625 Gb/s on each of 32 PCS lanes, which provides capacity for the MAC data rate of 800 Gb/s.

While this specification defines interfaces in terms of bits, octets, and frames, implementations may choose other data-path widths for implementation convenience.

172.1.5.1 PCS service interface

Change the text and insert Table 172-a in 172.1.5.1 as follows:

The PCS service interface allows the 800GBASE-R PCS to transfer information to and from a PCS client. The PCS client is either the Reconciliation Sublayer or the PHY 800GXS.

When the client sublayer is the Reconciliation Sublayer, the PCS service interface is the 800GMII defined in Clause 170.

When the client sublayer is the PHY 800GXS, the PCS service interface is the 800GMII with additional signals PCS_status, TXRD, TXLD, RXRD, RXLD, TXLS, and RXLS ~~PCS_status~~.

The PCS_status signal indicates the state of the PCS_status variable (see 119.2.6.2.2).

The TXRD, TXLD, RXRD, and RXLD signals support the FEC degrade function (see 172.1.4).

The TXRD and TXLD signals indicate the state of the remote degrade and local degrade, respectively, as detected by the PHY 800GXS in the transmit direction (see 171.3.3).

The RXRD signal indicates the state of the rx_rm_degraded variable (see 172.2.6.2.2) as detected by the PCS in the receive direction.

The RXLD signal is the logical OR of the FEC_degraded_SER and rx_local_degraded variables (see 172.2.6.2.2) as detected by the PCS in the receive direction.

~~The PCS_status signal indicates the state of the PCS_status variable (see 119.2.6.2.2) as detected by the PCS in the receive direction.~~

The TXLS and RXLS signals support passing link status information which would otherwise be handled by the IS_SIGNAL.request and IS_SIGNAL.indication primitives on an inter-sublayer service interface as described in 169.3. The TXLS signal indicates the state of the transmit link status as generated by the PHY 800GXS in the transmit direction (see 171.3.3). The RXLS signal takes the value of the rx_link_status variable in the receive direction.

The rx_link_status variable shall take one of the values of FAIL, IN_PROGRESS, READY, or OK as defined in Table 172-a.

Table 172-a—rx_link_status generation

<i>inst</i> :IS_SIGNAL.indication ^a SIGNAL_OK	align_status ^b	rx_link_status
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

^a From the sublayer below the PCS.

^b See 119.2.6.2.2

The TXLS and RXLS signals and rx_link_status variable are only required if the PCS is part of a PHY that supports autonomous path startup (APSU) (see Annex 178B).

Change the title and text of 172.1.5.2 and insert Table 172–b as follows:

172.1.5.2 Physical Medium Attachment (PMA) service interface ~~Service interface below the PCS~~

~~The PMA service interface below the PCS supports the exchange of encoded data between the PCS and PMA the sublayer below the PCS. The PMA service interface is defined in 173.3 and It is an instance of the inter-sublayer service interface definition defined in 169.3 and is described in an abstract manner which does not imply any particular implementation.~~

~~The PMA service interface for the PCS is described in an abstract manner and does not imply any particular implementation.~~

In the receive direction, the PCS receives 32 parallel streams of data via the inst:IS_UNITDATA_0:31.indication primitive and signal status information via the inst:IS_SIGNAL primitive.

In the transmit direction, the PCS transmits 32 parallel streams of data via the inst:IS_UNITDATA_0:31.request primitive and provides signal status information via the inst:IS_SIGNAL.request primitive. When the interface above the PCS is the Reconciliation Sublayer, the SIGNAL_OK parameter of the inst:IS_SIGNAL.request primitive is set to OK if the inst:IS_UNITDATA_0:31.request data units are valid and is set to FAIL otherwise. When the interface above the PCS is the PHY 800GXS, the SIGNAL_OK parameter at the interface below the PCS is set according to Table 172–b.

Table 172–b—inst:IS_SIGNAL.request(SIGNAL_OK) generation

TXLS	inst:IS_SIGNAL.request SIGNAL_OK
OK	OK
READY	READY
IN_PROGRESS	IN_PROGRESS
FAIL	FAIL

The inst:IS_SIGNAL.request primitive is only required if the PCS is part of a PHY that supports APSU (see Annex 178B).

172.1.6 Functional block diagram

Replace Figure 172–2 with the following figure:

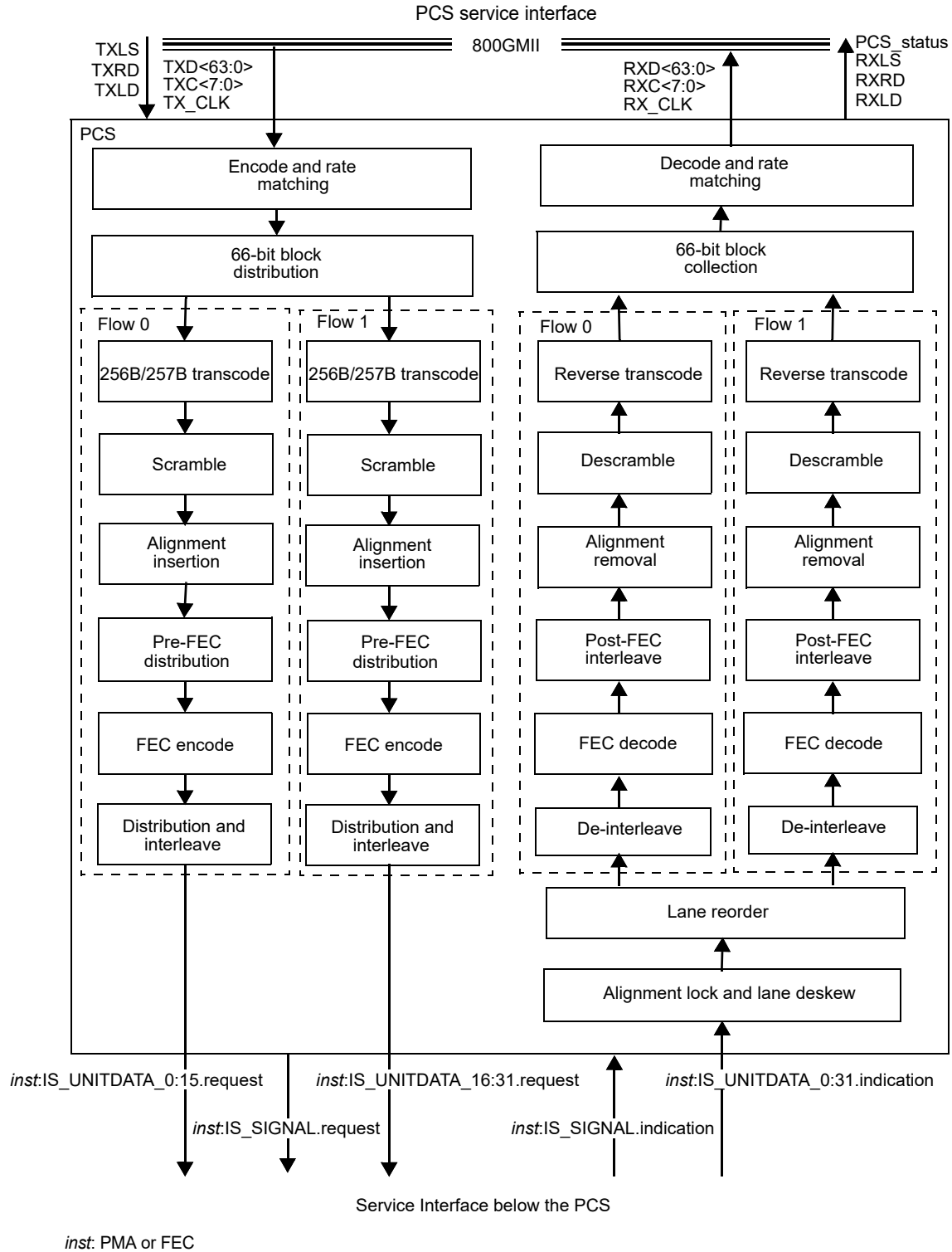


Figure 172–2—Functional block diagram

172.2 PCS functions

172.2.1 Overview

Change the text of 172.2.1 as follows:

The 800GBASE-R PCS is composed of the PCS Transmit and PCS Receive processes. The PCS shields the Reconciliation Sublayer (and MAC) from the specific nature of the underlying channel.

When communicating with the 800GMII, the PCS uses an eight octet-wide, synchronous data path, with packet delineation being provided by transmit control signals ($TXC_{<n>} = 1$) and receive control signals ($RXC_{<n>} = 1$). When communicating with the PMA sublayer below the PCS, the 800GBASE-R PCS uses 32 encoded bit streams also known as PCS lanes. Per direction (receive or transmit), the PCS lanes originate from a common clock but may vary in phase and skew dynamically. ~~The PMA operates independently of block and packet boundaries.~~ The PCS provides the functions necessary to map packets between the 800GMII ~~format~~ and the PMA service interface below the PCS ~~format~~.

The PCS transmit function can operate in normal mode or test-pattern mode.

When the transmit function is in normal mode, the PCS Transmit process continuously generates 66-bit blocks based on the $TXD_{<63:0>}$ and $TXC_{<7:0>}$ signals on the 800GMII. This takes place in the encode and rate matching function shown in Figure 172–2. The 66-bit blocks are then distributed between two parallel transmit functions, referred to as flow 0 and flow 1. Within each flow, the 66-bit blocks are transcoded to 257-bit blocks and scrambled. Alignment markers are periodically inserted into the data stream of each flow following the requirements specified in 172.2.4.6. For each flow, the data stream is distributed to two 5140-bit blocks and then FEC encoded. For each flow, the two FEC codewords are then interleaved before data is distributed to individual PCS lanes.

Transmit data units are sent to the service interface below the PCS via the PMA:IS_UNITDATA_0:31.request ~~inst:IS_UNITDATA_0:31.request~~ primitive.

When the transmit function is in test-pattern mode, a test pattern is packed into the transmit data units that are sent to the ~~PMA service interface~~ below the PCS via the PMA:IS_UNITDATA_0:31.request ~~inst:IS_UNITDATA_0:31.request~~ primitive.

In the receive direction, the PCS Synchronization process continuously monitors PMA:IS_SIGNAL.indication(SIGNAL_OK) ~~inst:IS_SIGNAL.indication(SIGNAL_OK)~~. When SIGNAL_OK indicates OK, then the PCS synchronization process accepts data units via the PMA:IS_UNITDATA_0:31.indication ~~inst:IS_UNITDATA_0:31.indication~~ primitive. It attains alignment marker lock based on the common marker (CM) portion of the alignment markers that are periodically transmitted on every PCS lane and identifies individual PCS lanes using the unique marker (UM) portion of the alignment marker. The PCS lanes are then reordered and deskewed, and the align_status flag is set. Note that a particular transmit PCS lane can be received on any receive lane of the service interface due to the skew and multiplexing that occurs in the path.

The PCS Receive process separates the reordered PCS lanes into two sets of 16 PCS lanes belonging to each flow. Within a flow, the data from the 16 PCS lanes is de-interleaved, processed by the FEC decoder, and re-interleaved on a 10-bit basis to form a single data stream. The alignment markers are removed, the data is descrambled and reverse transcoded back to 66-bit blocks. A 66-bit block collection function merges the 66-bit blocks from the two flows into a single stream of blocks that are then 64B/66B decoded.

172.2.4 Transmit function

172.2.4.1 64B/66B encoder

Change the text of 172.2.4.1 as follows:

The transmit PCS generates 66-bit blocks based on the TXD<63:0> and TXC<7:0> signals received from the 800GMII. Each 800GMII transfer is encoded into one 66-bit block. The contents of each 66-bit block are contained in a vector tx_coded<65:0> with tx_coded<1:0> containing the sync header and the remainder of the bits the payload. The transmit PCS shall use ~~the encoding method defined in either 172.2.4.1.1 or 172.2.4.1.2~~ one of two encoding methods, either the state-diagram method defined in 172.2.4.1.1 or a stateless method. If using a stateless method, then the stateless encoder defined in 119.2.4.1.2 should be used while the stateless encoder defined in 172.2.4.1.2 may be used.

172.2.4.2 Rate matching

Change the text of 172.2.4.2 as follows:

The transmit PCS may remove idle control characters or sequence ordered sets to compensate for the insertion of alignment markers. The transmit PCS may remove idle control characters or sequence ordered sets or may insert idle control characters to compensate for different clock domains on the 800GMII and the PMA service interface below the PCS. See 119.2.3.5 and 119.2.3.8 for the deletion and insertion rules.

172.2.4.10 Transmit bit ordering and distribution

Change the text of 172.2.4.10 as follows:

The transmit bit ordering and distribution is illustrated in Figure 172–4. The transmit bit ordering and distribution in “Flow 0 transmit function” and “Flow 1 transmit function” are identical to the 400GBASE-R transmit bit ordering and distribution in Figure 119–11 with the following exceptions:

- The “66-bit block distribution” of Figure 172–4 directly feeds into the “64B/66B to 256B/257B transcoder” in each flow’s transmit function shown in Figure 119–11, and the functional blocks above the “64B/66B to 256B/257B transcoder” in each flow are not used.
- For flow 1, the PMA service interface lane numbers 0:15 are replaced with 16:31, respectively.

Replace Figure 172–4 with the following figure:

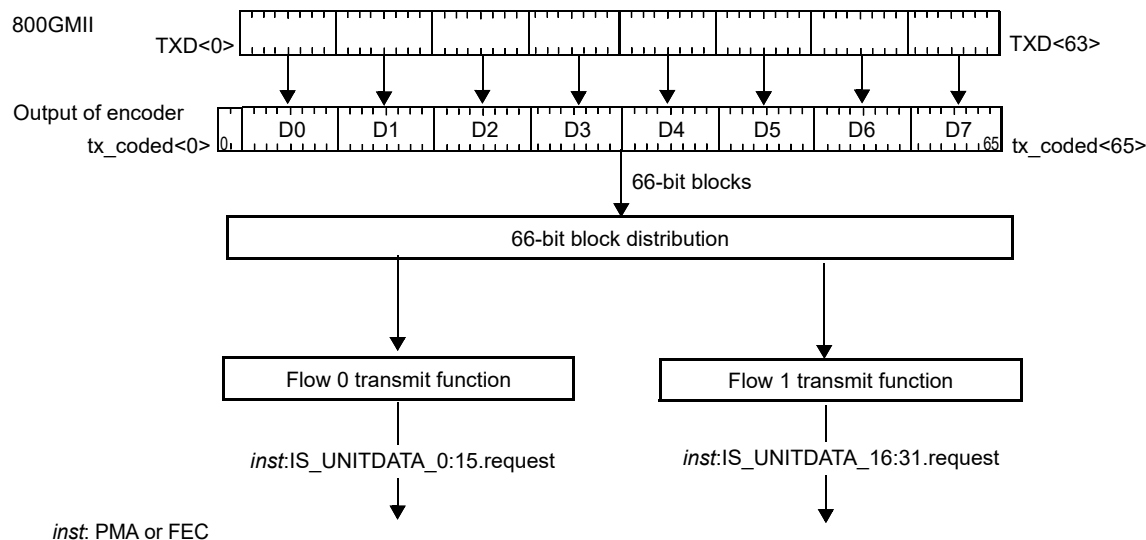


Figure 172–4—800GBASE-R PCS transmit bit ordering and distribution

172.2.5 Receive function

172.2.5.2 Lane reorder and de-interleave

Change the text of 172.2.5.2 as follows:

The PCS lanes might be received in any order on the PMA service interface lanes from the service interface below the PCS. The PCS receive function shall order the PCS lanes according to the PCS lane number. The PCS lane number is defined by the unique portion (UM₀ to UM₅) of the alignment marker that is mapped to each PCS lane (see 172.2.4.6).

After all 32 PCS lanes are aligned, deskewed, and reordered, they are separated into two sets of 16 PCS lanes belonging to each flow. PCS lanes 0 to 15 are sent to flow 0 and PCS lanes 16 to 31 are sent to flow 1. Within a flow, the data from the 16 PCS lanes is de-interleaved to reconstruct the original two streams of FEC codewords.

172.2.5.3 Reed-Solomon decoder

Change the text of 172.2.5.3 as follows:

The Reed-Solomon decoder is identical to that specified in 119.2.5.3, with the following exceptions:

- FEC_degraded_SER in flow 0 and flow 1 are mapped to FEC_degraded_SER_0 and FEC_degraded_SER_1, respectively.
- hi_ser in flow 0 and flow 1 are mapped to hi_ser_0 and hi_ser_1, respectively.

The FEC degrade detection is optional.

If bypass error indication is not supported or not enabled, when the Reed-Solomon decoder determines that a codeword contains errors that were not corrected, it causes the PCS receive function to set every 66-bit block within the two associated codewords to an error block (set to EBLOCK_R). When the stateless 64B/66B decoder specified in 172.2.5.9.2 is used, it is recommended that the first four 66-bit blocks (accounting for any alignment marker removal) following the two associated uncorrected codewords are also set to error blocks. If the stateless 64B/66B decoder specified in 119.2.5.8.2 is used, the setting of the following four 66-bit blocks to error blocks is required.

172.2.5.9 64B/66B decoder

Change the text of 172.2.5.9 as follows:

The receive PCS decodes 66-bit blocks to produce RXD<63:0> and RXC<7:0> for transmission to the 800GMII. One 800GMII transfer is decoded from each 66-bit block. The receive PCS shall use one of the two decoding methods that are defined in 172.2.5.9.1 and 172.2.5.9.2, a state-diagram based method as defined in 172.2.5.9.1 or a stateless method. If using a stateless method, the stateless decoder defined in 119.2.5.8.2 should be used while the stateless decoder defined in 172.2.5.9.2 may be used.

172.2.5.10 Rate matching

Change the text in 172.2.5.10 as follows:

The receive PCS may insert idle control characters to compensate for the deletion of alignment markers. The receive PCS may remove idle control characters or sequence ordered sets or may insert idle control characters to compensate for different clock domains on the PMA between the service interface below the PCS and the 800GMII. See 119.2.3.5 and 119.2.3.8 for the deletion and insertion rules.

NOTE—If no idle control characters are inserted to compensate for the removal of alignment markers, the average data rate on the 800GMII will be reduced by approximately 49 ppm.

172.3 PCS management

172.3.1 PCS MDIO function mapping

Change Table 172–6 as follows (some unchanged rows not shown):

Table 172–6—MDIO/PCS status variable mapping

MDIO status variable	PCS register name	Register/bit number	PCS status variable
...			
PCS FEC symbol errors, PCS lanes 0 to 31	PCS FEC symbol error counter register, lanes 0 to 31	3.600 to 3.663	FEC_symbol_error_counter<0:31>
<u>In-progress ability</u>	<u>PCS status 1</u>	<u>3.1.4</u>	<u>use_in_progress</u>
<u>In-progress indication</u>	<u>PCS status 1</u>	<u>3.1.3</u>	<u>in_progress_indication</u>

Table 172–6—MDIO/PCS status variable mapping (*continued*)

MDIO status variable	PCS register name	Register/bit number	PCS status variable
PCS FEC bypass indication ability	PCS FEC status register	3.801.1	FEC_bypass_indication_ability
...			

172.6 Auto-Negotiation

Change the text in 172.6 as follows:

The following requirements apply to a PCS used with an 800GBASE-CR8, ~~800GBASE-CR4, or~~ 800GBASE-KR8, ~~or 800GBASE-KR4~~ PMD where support for the Auto-Negotiation process defined in Clause 73 is mandatory. The PCS shall support the AN_LINK.indication(link_status) primitive (see 73.9.1). The parameter link_status shall take the value FAIL when PCS_status=false and the value OK when PCS_status=true one of the values FAIL, IN_PROGRESS, or OK, according to Table 172–6a. The primitive shall be generated when the value of link_status changes.

Insert new Table 172-6a as follows:

Table 172–6a—AN_LINK.indication(link_status) generation

reset + restart_lock	align_status	use_in_progress	link_status
true	don't care	don't care	FAIL
false	false	false	FAIL
		true	IN_PROGRESS
	true	don't care	OK

Insert the following new text in 172.6 below Table 172-6a as follows:

The PCS variables reset, restart_lock, and align_status are defined in [172.2.6.2.2](#).

The following PCS status variables are defined:

use_in_progress

Boolean variable indicating support of the value IN_PROGRESS for the link_status parameter of AN_LINK.indication. It is true for a PCS in the same package as a SM-PMA. Otherwise, its value is implementation dependent. If a Clause 45 MDIO is implemented, the status of this variable is reflected in bit 3.1.4

in_progress_indication

Boolean variable that is true when the link_status parameter of AN_LINK.indication has the value IN_PROGRESS and is false otherwise. If a Clause 45 MDIO is implemented, the status of this variable is reflected in bit 3.1.3

172.7 Protocol implementation conformance statement (PICS) proforma for Clause 172, Physical Coding Sublayer (PCS), type 800GBASE-R⁴

172.7.4 PICS proforma tables for Physical Coding Sublayer (PCS), type 800GBASE-R

172.7.4.1 Transmit function

Change the table in 172.7.4.1 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
*SE	Uses stateless encoder	172.2.4.1.2		$\emptyset$	Yes [] No []
TF1	Transmit 64B/66B encoder uses state-diagram encoder defined in Figure 119-14 172.2.4.1.1	172.2.4.1.1 172.2.4.1		SE:M O/I	Yes [] N/A []
TF2	Transmit 64B/66B encoder uses stateless rules encoder defined in Table 172-1 172.2.4.1.2	172.2.4.1.2 172.2.4.1		SE:M O/I	Yes [] N/A []
<u>TF3</u>	<u>Transmit 64B/66B encoder uses stateless encoder defined in 119.2.4.1.2</u>	172.2.4.1		O/I	<u>Yes []</u> <u>N/A []</u>
TF3 <u>4</u>	64B/66B to 256B/257B transcoder	172.2.4.4		M	Yes []
TF4 <u>5</u>	Scrambler	172.2.4.5		M	Yes []
TF5 <u>6</u>	Pad value	172.2.4.6		M	Yes []
TF6 <u>7</u>	Alignment marker insertion	172.2.4.6		M	Yes []
TF7 <u>8</u>	FEC degrade signaling	172.2.4.6		M	Yes []
TF8 <u>9</u>	Pre-FEC distribution	172.2.4.7		M	Yes []
TF9 <u>10</u>	Reed-Solomon encoder	172.2.4.8		M	Yes []
TF10 <u>1</u>	Symbol distribution	172.2.4.9		M	Yes []
TF11 <u>2</u>	Transmission bit ordering	172.2.4.10		M	Yes []
TF12 <u>3</u>	Scrambled idle transmit test-pattern generator	172.2.4.11		M	Yes []

⁴Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

172.7.4.2 Receive function

Change the table in 172.7.4.2 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
*BI	Bypass indication supported	172.2.5.3		O	Yes [] No []
*SD	Uses stateless decoder	172.2.5.9.2		O	Yes [] No []
RF1	Alignment marker lock	172.2.5.1		M	Yes []
...					
RF12	Receive 64B/66B decoder uses state-diagram decoder defined in Figure 119-15 172.2.5.9.1	172.2.5.9.1 172.2.5.9		SD:M O/2	Yes [] N/A []
RF13	Receive 64B/66B decoder uses stateless rules decoder defined in Table 172-4 172.2.5.9.2	172.2.5.9.2 172.2.5.9		SD:M O/2	Yes [] N/A []
<u>RF14</u>	<u>Receive 64B/66B decoder uses stateless decoder defined in 119.2.5.8.2</u>	<u>172.2.5.9</u>		<u>O/2</u>	

172.7.4.7 Auto-Negotiation

Change the table in 172.7.4.7 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
*AN	Support for use with an 800GBASE-CR8, or 800GBASE-CR4, 800GBASE-KR8, or PMD 800GBASE-KR4 PMD	172.6	AN technology dependent interface described in Clause 73	O	Yes [] No []
AN1	AN_LINK.indication primitive	172.6	Support of the AN_LINK.indication(link_status) primitive	AN:M	Yes [] N/A []
AN2	link_status parameter	172.6	Takes the value OK, <u>IN_PROGRESS</u> or FAIL	AN:M	Yes [] N/A []
AN3	Generation of AN_LINK.indication primitive	172.6	Generated when the value of link_status changes	AN:M	Yes [] N/A []